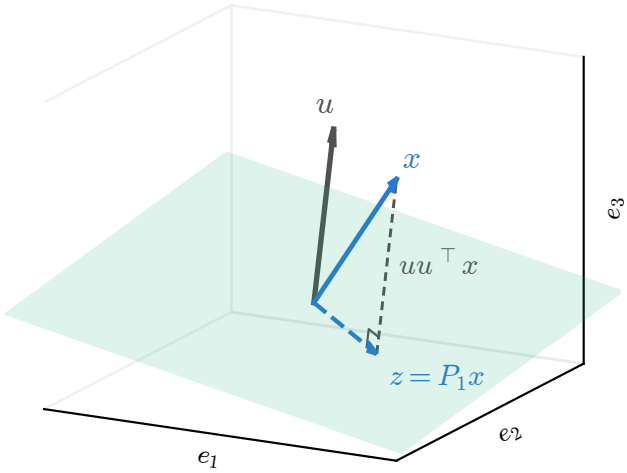
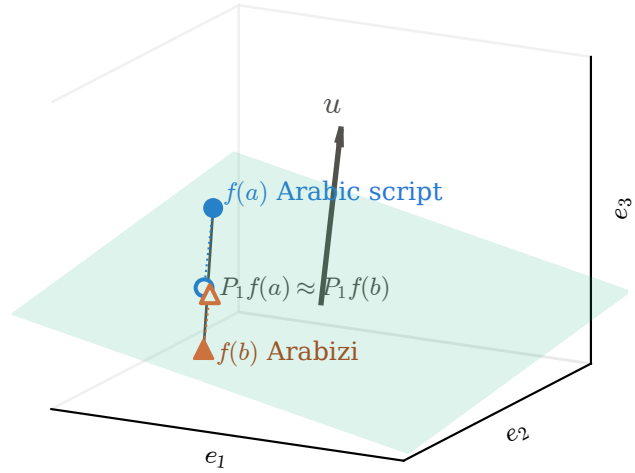
(a) $x = P_1 x + uu^\top x$



shaded plane $= \mathrm{span}(u)^\perp = \mathrm{range}(P_1)$ (retained)

(b) a pair before and after P_1



$\|\Delta\| = 1.05 \quad \rightarrow \quad \|P_1 \Delta\| = 0.10$

Schematic: constructed vectors in $\mathbb{R}^3$, not measured embeddings